PH 410 Course coordinator/ Instructor**: Alika Khare**
Problem Set no. VII **Due date: Wednesday, Mar 19, 2025**

Note: . Please submit the solutions of at least two of the problems, on Wednesday, Mar 19, 2025, during the class. The solutions can be submitted in a group of maximum of four students. Name and Roll no. of all the group members should be written clearly on the top. If solutions are found to be reasonably correct then 10 marks (5marks each) will be awarded and in case of non submission/wrong solution zero marks.

1. A point charge is embedded at the center of the dielectric sphere of radius R and susceptibility χ_e. Calculate the polarization, all the bound charge densities and electric field everywhere. Give a tentative sketch of electric field as a function of r (distance from the center of the sphere).
2. A point electric dipole is embedded at the center of a dielectric sphere of radius *R* and dielectric constant *k*. Find the electric potential and hence the electric field.
3. An uncharged conducting sphere of radius *a* is coated with a spherical shell of dielectric material inner and outer radius *a* and *b* respectively. This system is placed in an otherwise uniform electric field of very large extent. Find the electric field within the dielectric medium and obtain all the possible bound charge densities.
4. Obtained the polarizability of a gaseous atom assuming the classical model of an atom where the positively charged nucleus at the center is surrounded by uniformly distributed electronic cloud.
5. A very long cylinder of linear dielectric material is placed in an otherwise uniform electric field; the axis of the cylinder is perpendicular to the electric field. Find the potential and hence the electric field inside the cylinder.

6. Problems from D J Griffith IV th edition: 4.4, 4.10, 4.15, 4.33, 4.34.

PH410: ELECTRODYNAMICS-I
Tutorial 7

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1. A point charge is embedded at the center of the dielectric sphere of radius R and susceptibility χ_e. Calculate the polarization, all the bound charge densities and electric field everywhere. Give a tentative sketch of electric field as a function of r (distance from the center of the sphere).

2. A point electric dipole is embedded at the center of a dielectric sphere of radius R and dielectric constant k. Find the electric potential and hence the electric field.

3. An uncharged conducting sphere of radius a is coated with a spherical shell of dielectric material inner and outer radius a and b respectively. This system is placed in an otherwise uniform electric field of very large extent. Find the electric field within the dielectric medium and obtain all the possible bound charge densities.

4. Obtained the polarizability of a gaseous atom assuming the classical model of an atom where the positively charged nucleus at the center is surrounded by uniformly distributed electronic cloud.

5. A very long cylinder of linear dielectric material is placed in an otherwise uniform electric field; the axis of the cylinder is perpendicular to the electric field. Find the potential and hence the electric field inside the cylinder

Probem 4.4 A point charge q is situated a large distance r from a neutral atom of polarizability α. Find the force of attraction between them.

Problem 4.10 A sphere of radius R carries a polarization

$$\mathbf{P}(\mathbf{r}) = k\mathbf{r}, \tag{1}$$

where k is a constant and $\mathbf{r}$ is the vector from the center.

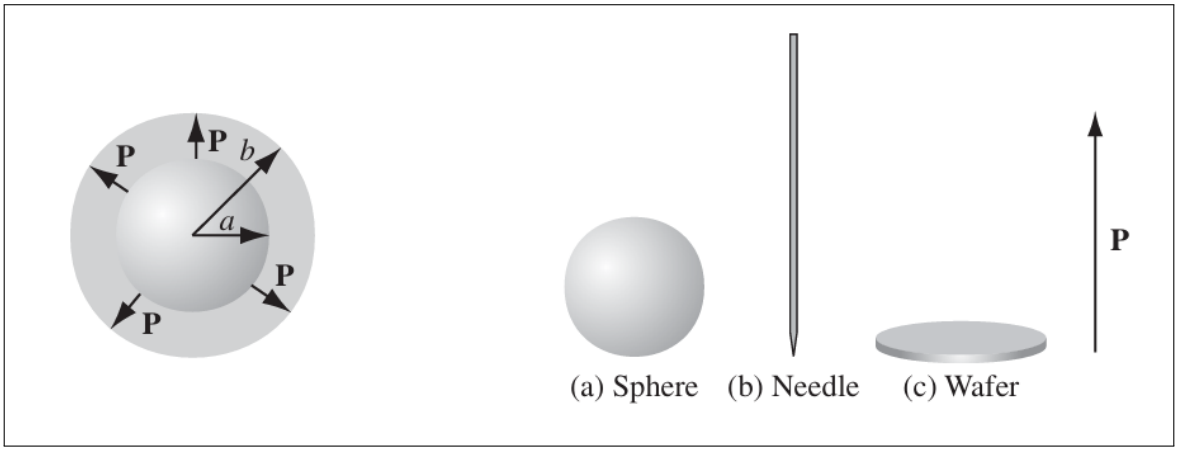
(a) Calculate the bound charges σ_b and ρ_b.

(b) Find the field inside and outside the sphere.

Problem 4.15 A thick spherical shell (inner radius a, outer radius b) is made of dielectric material with a "frozen-in" polarization

$$\mathbf{P}(\mathbf{r}) = \frac{k}{r}\hat{\mathbf{r}}, \tag{2}$$

where k is a constant and r is the distance from the center (Fig. 4.18). (There is no *free* charge in the problem.) Find the electric field in all three regions by two different methods:



(a) Locate all the bound charge, and use Gauss's law (Eq. 2.13) to calculate the field it produces.

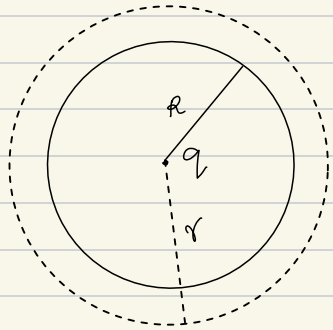
(b) Use Eq. 4.23 to find **D**, and then get **E** from Eq. 4.21. [Notice that the second method is much faster, and it avoids any explicit reference to the bound charges.]

Problem 4.33 A dielectric cube of side a, centered at the origin, carries a "frozen-in" polarization

$$\mathbf{P} = k\mathbf{r}, \tag{3}$$

where k is a constant. Find all the bound charges, and check that they add up to zero.

1. A point charge is embedded at the center of the dielectric sphere of radius R and susceptibility χ_e. Calculate the polarization, all the bound charge densities and electric field everywhere. Give a tentative sketch of electric field as a function of r (distance from the center of the sphere).



Using Gauss law

$$\oint D \cdot da = Q_{enc}$$

Or $D \cdot 4\pi r^2 = q \Rightarrow \vec{D} = \frac{q}{4\pi r^2}\hat{r}$

The electric field inside is therefore given by

$$\boxed{\frac{\epsilon}{\epsilon_0} = K = 1 + \chi_e}$$

$$\vec{E} = \frac{q}{4\pi\epsilon r^2}\hat{r} = \frac{q}{4\pi\epsilon_0(1+\chi_e)r^2}\hat{r} \quad (r < R)$$

$$\vec{P} = \epsilon_0 \chi_e \vec{E} = \frac{q\chi_e}{4\pi(1+\chi_e)r^2}\hat{r}$$

Bound vol. charge density is

$$\rho_b = -\vec{\nabla}\cdot\vec{P} = -\frac{q\chi_e}{4\pi(1+\chi_e)}\vec{\nabla}\cdot\left(\frac{\hat{r}}{r^2}\right) = -\frac{q\chi_e}{4\pi(1+\chi_e)}\delta^3(r)$$

Surface bound charge density is

$$\sigma_b = \vec{P}\cdot\hat{n} = \frac{q\chi_e}{4\pi(1+\chi_e)R^2}\hat{r}\cdot\hat{r} = \frac{q\chi_e}{4\pi(1+\chi_e)R^2}$$

Now, the total bound surface charge is

$$Q_{surface} = \oint \sigma_b \, da = \sigma_b \cdot 4\pi R^2 = \frac{q\chi_e}{(1+\chi_e)}$$

The total bound volume charge is

$$Q_{volume} = \oint \rho_b \, d\tau = -\frac{q\chi_e}{4\pi(1+\chi_e)}\int \delta^3(r)\, d\tau = \frac{-q\chi_e}{(1+\chi_e)}$$

2. A point electric dipole is embedded at the center of a dielectric sphere of radius R and dielectric constant k. Find the electric potential and hence the electric field.

The general soln. for laplace eqn is

$$V(r,\theta) = \sum_{l=0}^{\infty}\left(A_l r^l + \frac{B_l}{r^{l+1}}\right) P_l(\cos\theta)$$

Inside: $V_1(r,\theta) = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{Kr^2} + \sum_{l=0}^{\infty} A_l r^l P_l(\cos\theta) \qquad r \leq R$

Outside: $V_2(r,\theta) = \sum_{l=0}^{\infty} \frac{B_l}{r^{l+1}} P_l(\cos\theta) \qquad r \geq R$

Boundary conditions:

i) V is continuous at $r = R$

ii) $D_1^{\perp} = D_2^{\perp}$ at $r = R$

Using the first boundary condition

$$\frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{KR^2} + \sum_{l=0}^{\infty} A_l R^l P_l(\cos\theta) = \sum_{l=0}^{\infty}\frac{B_l}{R^{l+1}} P_l(\cos\theta)$$

For $l = 1$

$$\frac{p}{4\pi\varepsilon_0 K R^2} + A_1 R = \frac{B_1}{R^2}$$

or $$B_1 = \frac{p}{4\pi\varepsilon_0 K} + A_1 R^3 \quad \text{——} \quad (1)$$

For $l \neq 1$

$$A_l R^l = \frac{B_l}{R^{l+1}} \Rightarrow B_l = A_l R^{2l+1} \quad \text{——} \quad (2)$$

Also from the second boundary condition

$$\varepsilon \frac{\partial V_1}{\partial r}\Big|_{r=R} = \varepsilon_0 \frac{\partial V_2}{\partial r}\Big|_{r=R} =$$

$$\varepsilon\left(\sum l A_l R^{l-1} P_l(\cos\theta) - \frac{1}{4\pi\varepsilon_0}\frac{2p\cos\theta}{KR^3}\right) = -\varepsilon_0 \sum (l+1)\frac{B_l}{R^{l+2}} P_l(\cos\theta)$$

$$K \sum l A_l R^{l-1} P_l(\cos\theta) - \frac{1}{4\pi\varepsilon_0} \frac{2p\cos\theta}{R^3} = -\sum (l+1) \frac{B_l}{R^{l+2}} P_l(\cos\theta)$$

For $l = 1$,

$$K A_1 - \frac{1}{4\pi\varepsilon_0} \frac{2p}{R^3} = -2 \frac{B_1}{R^3}$$

$$B_1 = \frac{p}{4\pi\varepsilon_0} - \frac{K R^3 A_1}{2} \quad \text{——— (3)}$$

For $l \neq 1$

$$K l A_l R^{l-1} = -(l+1) \frac{B_l}{R^{l+2}}$$

$$B_l = -\frac{K l}{(l+1)} A_l R^{2l+1} \quad \text{——— (4)}$$

From (1) & (3)

$$\frac{p}{4\pi\varepsilon_0 K} + A_1 R^3 = \frac{p}{4\pi\varepsilon_0} - \frac{K R^3 A_1}{2}$$

$$A_1 R^3 \left(1 + \frac{K}{2}\right) = \frac{p}{4\pi\varepsilon_0} \left(1 - \frac{1}{K}\right)$$

$$A_1 R^3 \left(\frac{2+K}{2}\right) = \frac{p}{4\pi\varepsilon_0} \left(\frac{K-1}{K}\right)$$

$$A_1 = \frac{p}{4\pi\varepsilon_0 R^3} \frac{2(K-1)}{K(K+2)} \quad \text{——— (5)}$$

From (1) & (5)

$$B_1 = \frac{p}{4\pi\varepsilon_0 K} + \frac{p}{4\pi\varepsilon_0 R^3} \frac{2(K-1)}{K(K+2)} R^3$$

$$B_1 = \frac{p}{4\pi\varepsilon_0} \left[\frac{1}{K} + \frac{2(K-1)}{K(K+2)}\right] = \frac{p}{4\pi\varepsilon_0} \frac{3K}{K(K+2)}$$

Also from (2) & (4) both eqn satisfies only if $A_l = B_l = 0$

Then the potential is

$$V_1(r,\theta) = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{Kr^2} + \frac{p}{4\pi\varepsilon_0 R^3}\frac{2(K-1)}{K(K+2)}r\cos\theta$$

$$= \frac{p\cos\theta}{4\pi\varepsilon_0 r^2 K}\left[1 + \frac{2(K-1)}{K(K+2)}\frac{r^3}{R^3}\right] //$$

$$V_2(r,\theta) = \frac{p}{4\pi\varepsilon_0}\frac{3K}{K(K+2)}\cdot\frac{1}{r^2}\cos\theta$$

$$= \frac{p\cos\theta}{4\pi\varepsilon_0 r^2}\frac{3K}{K(K+2)} //$$

Electric fields are:

$$E = -\nabla V \Rightarrow \nabla V = \frac{\partial V}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial V}{\partial \theta}\hat{\theta}$$

Inside:

$$E_r = -\frac{\partial V}{\partial r} = \frac{2p\cos\theta}{4\pi\varepsilon_0 K r^3} - \frac{6p(K-1)\cos\theta}{4\pi\varepsilon_0 K(K+2)R^3}r^2$$

$$E_\theta = -\frac{\partial V}{\partial \theta} = \frac{p\sin\theta}{4\pi\varepsilon_0 K r^3}\left[1 + 2\frac{(K-1)}{(K+2)}\frac{r^3}{R^3}\right]$$

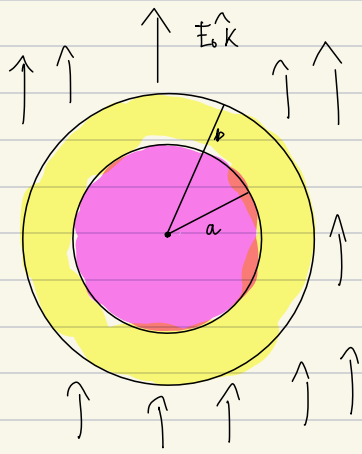
Outside:

$$E_r = \frac{6p\cos\theta}{4\pi\varepsilon_0(K+2)r^3}; \quad E_\theta = -\frac{3p\sin\theta}{4\pi\varepsilon_0(K+2)r^3}$$

Therefore,

$$\vec{E} = \begin{cases} \frac{2p\cos\theta}{4\pi\varepsilon_0 K r^3}\left(1 - \frac{3(K-1)}{(K+2)}\frac{r^5}{R^3}\right)\hat{r} + \frac{p\sin\theta}{4\pi\varepsilon_0 K r^3}\left[1 + \frac{2(K-1)r^3}{(K+2)R}\right]\hat{\theta}, & r \leq R \\ \frac{3p}{4\pi\varepsilon_0(K+2)r^3}(3\cos\theta\,\hat{r} - \sin\theta\,\hat{\theta}), & r \geq R \end{cases}$$

3. An uncharged conducting sphere of radius a is coated with a spherical shell of dielectric material inner and outer radius *a* and *b* respectively. This system is placed in an otherwise uniform electric field of very large extent. Find the electric field within the dielectric medium and obtain all the possible bound charge densities.



$V_{in}(r,\theta) = 0$ (conductor)

$$V_{med}(r,\theta) = \sum \left(A_\ell r^\ell + \frac{B'_\ell}{r^{\ell+1}} \right) P_\ell(\cos\theta)$$

$$V_{out}(r,\theta) = -E_0 r\cos\theta + \sum \frac{B_\ell}{r^{\ell+1}} P_\ell(\cos\theta)$$

Boundary conditions are

i) $V_{in} = V_{med}$ at $r = a$

ii) $V_{med} = V_{out}$, at $r = b$

iii) $D^\perp_{med} = D^\perp_{out}$ at $r = b$

From the first boundary condition

$$\sum \left(A_\ell a^\ell + \frac{B'_\ell}{a^{\ell+1}} \right) P_\ell(\cos\theta) = 0$$

$$B'_\ell = -A_\ell a^{2\ell+1} \quad \text{——— (1)}$$

From the 2nd Boundary condition

$$\sum \left(A_\ell b^\ell + \frac{B'_\ell}{b^{\ell+1}} \right) P_\ell(\cos\theta) = -E_0 b\cos\theta + \sum \frac{B_\ell}{b^{\ell+1}} P_\ell(\cos\theta)$$

For $\ell = 1$

$$A_1 + \frac{B'_1}{b^2} = -E_0 b + \frac{B_1}{b^2}$$

Using (1)

$$A_1 - \frac{A_1 a^3}{b^2} = -E_0 b + \frac{B_1}{b^2}$$

$$B_1 = A_1(b^3 - a^3) + E_0 b^3 \quad \text{——— (2)}$$

For $l \neq 1$

$$A_l b^l + \frac{B'_l}{b^{l+1}} = \frac{B_l}{b^{l+1}}$$

$$A_l b^l - \frac{A_l a^{2l+1}}{b^{l+1}} = \frac{B_l}{b^{l+1}} \quad (\text{using } ①)$$

$$B_l = A_l (b^{2l+1} - a^{2l+1}) \quad \text{——} \quad ③$$

From third boundary condition

$$\varepsilon \frac{\partial V_{med}}{\partial r}\Big|_{r=b} = \varepsilon_0 \frac{\partial V_{out}}{\partial r}\Big|_{r=b}$$

$$K \sum \left(l A_l b^{l-1} - (l+1) \frac{B'_l}{b^{l+2}} \right) P_l(\cos\theta) = -E_0 \cos\theta - \sum (l+1) \frac{B_l}{b^{l+2}} P_1(\cos\theta)$$

For $l = 1$

$$K \left(A_1 + \frac{2 a^3 A_1}{b^3} \right) = -E_0 - 2 \frac{B_1}{b^3}$$

$$-2 B_1 - E_0 b^3 = K A_1 (b^3 + 2a^3) \quad \text{——} \quad ④$$

Using ②

$$-2 \left[A_1 (b^3 - a^3) + E_0 b^3 \right] - E_0 b^3 = K A_1 (b^3 + 2a^3)$$

$$-2 A_1 (b^3 - a^3) - 3 E_0 b^3 = K A_1 (b^3 + 2a^3)$$

$$A_1 = \frac{-3 E_0 b^3}{2[b^3 - a^3] + K(b^3 + 2a^3)} = \frac{-3 E_0}{2(1 - (a/b)^3) + K(1 + 2(a/b)^3)} \quad \text{——} \quad ⑤$$

For $l \neq 1$

$$K \left[l A_l b^{l-1} + (l+1) \frac{B'_l}{b^{l+2}} \right] = -(l+1) \frac{B_l}{b^{l+2}}$$

$$K \left[l A_l b^{l-1} + (l+1) \frac{a^{2l+1} A_l}{b^{l+2}} \right] = -(l+1) \frac{B_l}{b^{l+2}}$$

$$B_l = -K A_l \left[\frac{l}{(l+1)} b^{2l+1} + a^{2l+1} \right] \quad \text{——— (6)}$$

For (3) and (6) to be true B_l & A_l must equal zero

Using (1) & (5)

$$B_1' = \frac{-3E_0 a^3}{2(1-(a/b)^3) + K(1+2(a/b)^3)} \quad \text{——— (7)}$$

The potential for the medium is

$$V_{med}(r,\theta) = \frac{-3E_0 a^3 r\cos\theta}{2(1-(a/b)^3) + K(1+2(a/b)^3)} + \frac{-3E_0 a^3}{2(1-(a/b)^3) + K(1+2(a/b)^3)} \cdot \frac{1}{r^2}\cos\theta$$

$$= \frac{-3E_0 a^3}{2(1-(a/b)^3) + K(1+2(a/b)^3)} \left(r - \frac{1}{r^2} \right) \cos\theta$$

Electric field is

$$E_r = -\frac{\partial V}{\partial r} = \frac{3E_0}{2(1-(a/b)^3) + K(1+2(a/b)^3)} \left(1 + \frac{2a^3}{r^3} \right) \cos\theta$$

$$E_\theta = -\frac{1}{r}\frac{\partial V}{\partial \theta} = \frac{-3E_0}{2(1-(a/b)^3) + K(1+2(a/b)^3)} \left(1 - \frac{a^3}{r^3} \right) \sin\theta$$

$$\therefore \vec{E} = \frac{3E_0}{2(1-(a/b)^3) + K(1+2(a/b)^3)} \left\{ \left(1 + \frac{2a^3}{r^3} \right) \cos\theta\,\hat{r} + \left(1 - \frac{a^3}{r^3} \right) \sin\theta\,\hat{\theta} \right\}$$

Now

$$\vec{P} = \varepsilon_0 \chi_e \vec{E}$$

$$= \frac{3E_0\varepsilon_0(K-1)}{2(1-(a/b)^3) + K(1+2(a/b)^3)} \left\{ \left(1 + \frac{2a^3}{r^3} \right) \cos\theta\,\hat{r} + \left(1 - \frac{a^3}{r^3} \right) \sin\theta\,\hat{\theta} \right\}$$

$$\sigma_b = P.\hat{n}$$

At $r = a$

$$\sigma_b = \frac{3 E_0 \varepsilon_0 (k-1)}{2(1-(a/b)^3) + k(1+2(a/b)^3)} \left(1 + \frac{2a^3}{a^3}\right) \cos\theta$$

$$= \frac{9 E_0 \varepsilon_0 (k-1)}{2(1-(a/b)^3) + k(1+2(a/b)^3)} \cos\theta$$

At $r = b$

$$\sigma_b = \frac{3 E_0 \varepsilon_0 (k-1)}{2(1-(a/b)^3) + k(1+2(a/b)^3)} \left(1 + \frac{2a^3}{b^3}\right) \cos\theta$$

$$\rho_b = -\vec{\nabla}.\vec{P}$$

$$= -\left[\frac{1}{r^2}\frac{\partial}{\partial r}(r^2 P_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(P_\theta \sin\theta)\right]$$

$$\frac{1}{r^2}\frac{\partial}{\partial r}(r^2 P_r) = \frac{3 E_0 \varepsilon_0 (k-1)}{2(1-(a/b)^3) + k(1+2(a/b)^3)} \left(\frac{2}{r} - \frac{2a^3}{r^4}\right) \cos\theta$$

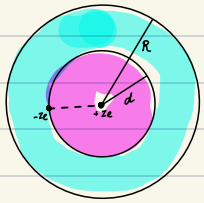
$$\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(P_\theta \sin\theta) = \frac{-3 E_0 \varepsilon_0 (k-1)}{2(1-(a/b)^3) + k(1+2(a/b)^3)} \left(\frac{2}{r}\left(1 - \frac{a^3}{r^3}\right)\cos\theta\right)$$

$$\therefore \rho_b = \frac{-3 E_0 \varepsilon_0 (k-1)}{2(1-(a/b)^3) + k(1+2(a/b)^3)} \left[\frac{2}{r} - \frac{2a^3}{r^4} - \frac{2}{r} + \frac{2a^3}{r^4}\right] \cos\theta$$

$$= 0$$

4. Obtained the polarizability of a gaseous atom assuming the classical model of an atom where the positively charged nucleus at the center is surrounded by uniformly distributed electronic cloud.

The induced dipole moment is



$$p = Zed$$

Now, we will consider a uniformly charged sphere with radius R with charge density

$$\rho = \frac{-Ze}{\frac{4}{3}\pi R^3}$$

According to Gauss law, Electric field inside a sphere at d

$$E_{restore} = \frac{Q_{enc}}{4\pi\varepsilon_0 d^2}$$

$$Q_{enc} = \rho \times \frac{4}{3}\pi d^3 = -Ze\left(\frac{d}{R}\right)^3$$

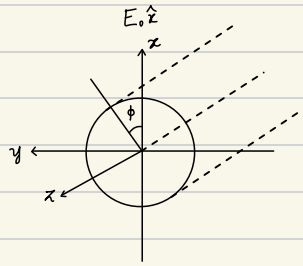
Then $E_{restore} = \dfrac{(-Ze)d}{4\pi\varepsilon_0 R^3}$

$$ZeE = \frac{(Ze)^2 d}{4\pi\varepsilon_0 R^3} \implies d = \frac{4\pi\varepsilon_0 R^3 E}{Ze}$$

$$p = Ze\left(\frac{4\pi\varepsilon_0 R^3 E}{Ze}\right) = 4\pi\varepsilon_0 R^3 E$$

$$\alpha = 4\pi\varepsilon_0 R^3$$

5. A very long cylinder of linear dielectric material is placed in an otherwise uniform electric field; the axis of the cylinder is perpendicular to the electric field. Find the potential and hence the electric field inside the cylinder



Boundary conditions:

1) $V_{in} = V_{out}$ at $\rho = R$

2) $D_1^{\perp} = D_2^{\perp}$ at $\rho = R$

3) $V_{out} \longrightarrow -E_0 \rho \cos\phi$ for $\rho \gg R$

The general solution is

$$V(\rho, \phi) = A_0 + B_0 \ln \rho + \sum_{n=1}^{\infty} \left(A_n \rho^n + \frac{B_n}{\rho^n} \right) (C_n \cos n\phi + D_n \sin n\phi)$$

$$V_{in}(\rho, \phi) = \sum_{n=1}^{\infty} \rho^n (a_n \cos n\phi + b_n \sin n\phi)$$

$$V_{out}(\rho, \phi) = -E_0 \rho \cos\phi + \sum_{n=1}^{\infty} \rho^{-n} (c_n \cos n\phi + d_n \sin n\phi)$$

Using first Boundary condition

$$\sum_{n=1}^{\infty} R^n (a_n \cos n\phi + b_n \sin n\phi) = -E_0 R \cos\phi + \sum_{n=1}^{\infty} R^{-n} (c_n \cos n\phi + d_n \sin n\phi)$$

$$R^n a_n = -E_0 R + R^{-n} c_n \quad \text{——— (1)}$$

$$R^n b_n = R^{-n} d_n \Rightarrow d_n = R^{-2n} b_n \quad \text{——— (2)}$$

From the 2nd Boundary condition

$$K \sum n R^{n-1} (a_n \cos n\phi + b_n \sin n\phi) = -E_0 \cos\phi - \sum n R^{-n-1} (c_n \cos n\phi + d_n \sin n\phi)$$

For $n \neq 1$

$$K n R^{n-1} a_n = -n R^{-n-1} c_n \quad \text{——— (3)}$$

$$K n R^{n-1} b_n = -n R^{-n-1} d_n \Rightarrow d_n = -K R^{2n} b_n \quad \text{——— (4)}$$

(2) & (4) are only true if all $d_n = b_n = 0$

This is due to the symmetry problem. The field is along x-axis which mean the potential due to this field should also be symmetric

Therefore, the flip from ϕ to $-\phi$ should give us same value but this is possible only if the sine function (odd function) is zero.

Also from (2) & (3) $a_n = c_n = 0$ for $n \neq 1$

For $n = 1$

$$K a_1 = -E_0 - R^{-2} C_1 \quad \text{——} \quad (5)$$

$$K(-E_0 + R^{-2} C_1) = -E_0 - R^{-2} C_1 \quad (\text{using } (1))$$

$$-KE_0 + KR^{-2} C_1 = -E_0 - R^{-2} C_1$$

$$C_1 R^{-2}(K+1) = E_0(K-1)$$

$$C_1 = \frac{E_0 R^2 \chi_e}{(\chi_e + 2)} \quad \text{——} \quad (6)$$

Using (1) & (6)

$$a_1 = -E_0 + R^{-2} \frac{E_0 R^2 \chi_e}{(\chi_e + 2)}$$

$$a_1 = -E_0 \left[\frac{\chi_e + 2 - \chi_e}{(\chi_e + 2)} \right]$$

$$a_1 = \frac{-2E_0}{(\chi_e + 2)} \quad \text{——} \quad (7)$$

$\therefore$ The potentials are

$$V_{in}(\rho, \phi) = \frac{-2E_0 \rho}{(\chi_e + 2)} \cos\phi = \frac{-2E_0}{(\chi_e + 2)} x$$

$$V_{out}(\rho, \phi) = -E_0 \rho \cos\phi + \frac{E_0 R^2 \chi_e}{(\chi_e + 2)} \rho^{-1} \cos\phi$$

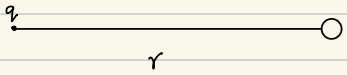
$$= -E_0 \rho \left[1 + \frac{\chi_e}{(\chi_e + 2)} \frac{R^2}{\rho^2} \right] \cos\phi$$

$$\vec{E}_{in} = -\vec{\nabla} \cdot \vec{V}_{in} = -\frac{d}{dx} \left[-\frac{2E_0}{(\chi_e + 2)} x \right]$$

$$= -\frac{2E_0}{(\chi_e + 2)} \hat{x}$$

[Question only ask for Inside potential, I just did the extra work here]

Probem 4.4 A point charge q is situated a large distance r from a neutral atom of polarizability α. Find the force of attraction between them.



Electric field due to q is

$$\vec{E} = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{r} \hat{r}$$

Polarization due to $\vec{E}$

$$\vec{P} = \alpha \vec{E}$$

$$= \frac{\alpha q}{4\pi\varepsilon_0 r} \hat{r}$$

Electric field to dipole is

$$\vec{E} = \frac{p}{4\pi\varepsilon_0 r^3} \left(2\cos\theta \hat{r} + \sin\theta \hat{\theta}\right)$$

At $\theta = \pi$

$$\vec{E} = \frac{-2p}{4\pi\varepsilon_0 r^3} \hat{r} = -\frac{1}{4\pi\varepsilon_0 r^3} \left(\frac{2\alpha q}{4\pi\varepsilon_0 r^2}\right) \hat{r}$$

$$\vec{F} = -2\alpha \left(\frac{q}{4\pi\varepsilon_0}\right)^2 \frac{1}{r^5} \hat{r}$$

(Attractive in nature)

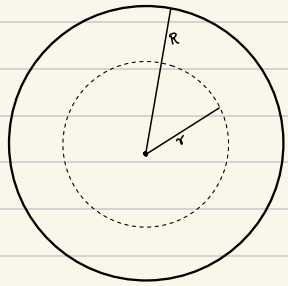
Problem 4.10 A sphere of radius R carries a polarization

$$\mathbf{P}(\mathbf{r}) = k\mathbf{r},$$

where k is a constant and $\mathbf{r}$ is the vector from the center.

(a) Calculate the bound charges σ_b and ρ_b.

(b) Find the field inside and outside the sphere.



$$P(r) = kr$$

a) $\sigma_b = \vec{P}\cdot\hat{n} = kR$

$$\rho_b = -\vec{\nabla}\cdot\vec{P} = -\frac{1}{r^2}\frac{d}{dr}\left(r^2 \cdot kr\right) = -3k$$

b) Inside:

$$\rho = \frac{q}{V}$$

Using Gauss law

$$\oint E\cdot dS = \frac{Q_{enc}}{\varepsilon_0} \Rightarrow E\cdot 4\pi r^2 = \frac{1}{\varepsilon_0}\rho_b \frac{4}{3}\pi r^3$$

$$E = -\frac{k}{\varepsilon_0} r$$

Outside:

$$\oint E\cdot dS = \frac{Q_{enc}}{\varepsilon_0} \Rightarrow E\cdot 4\pi r = \frac{1}{\varepsilon_0}\left[(-3k)\frac{4}{3}\pi R^3 + (kR)4\pi R^2\right]$$

$$E = 0$$